

الجامعة الألمانية الأردنية
German Jordanian University

Power Electronics Pre-Lab

DC – DC Converters (Buck-Boost)

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Buck

Parameter	Symbol	Value
Input Voltage	V_in	15 V
Switching Frequency	f_sw	5 kHz
Duty Cycle	D	40%
Load Resistance	R	50 Ω
Inductor	L	12.5 mH
Capacitor	C	16 µF
Average Output Voltage	V_dc	**6.0 V** *(Ideal)* / **~5.3 V** *(mean)*
Average Output Current	I_dc	**0.12 A** *(Ideal)* / **~0.106 A** *(mean)*

Circuit Diagram

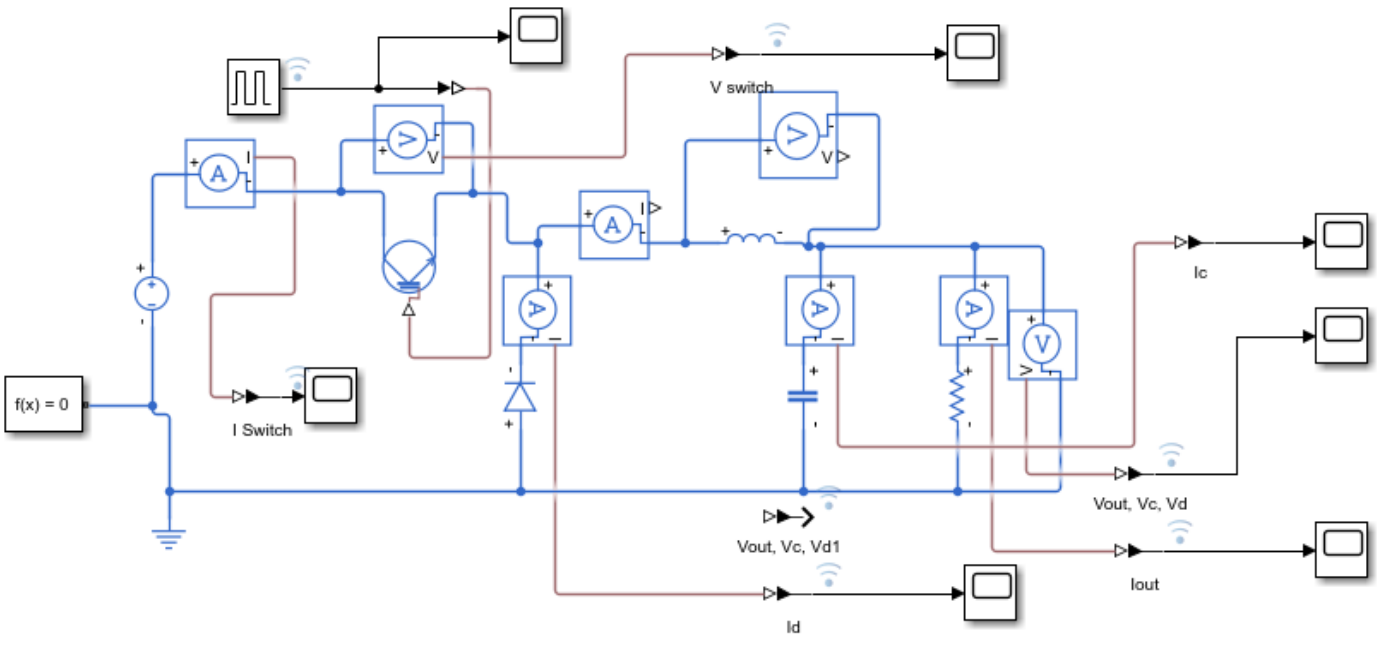


Figure 1- Buck

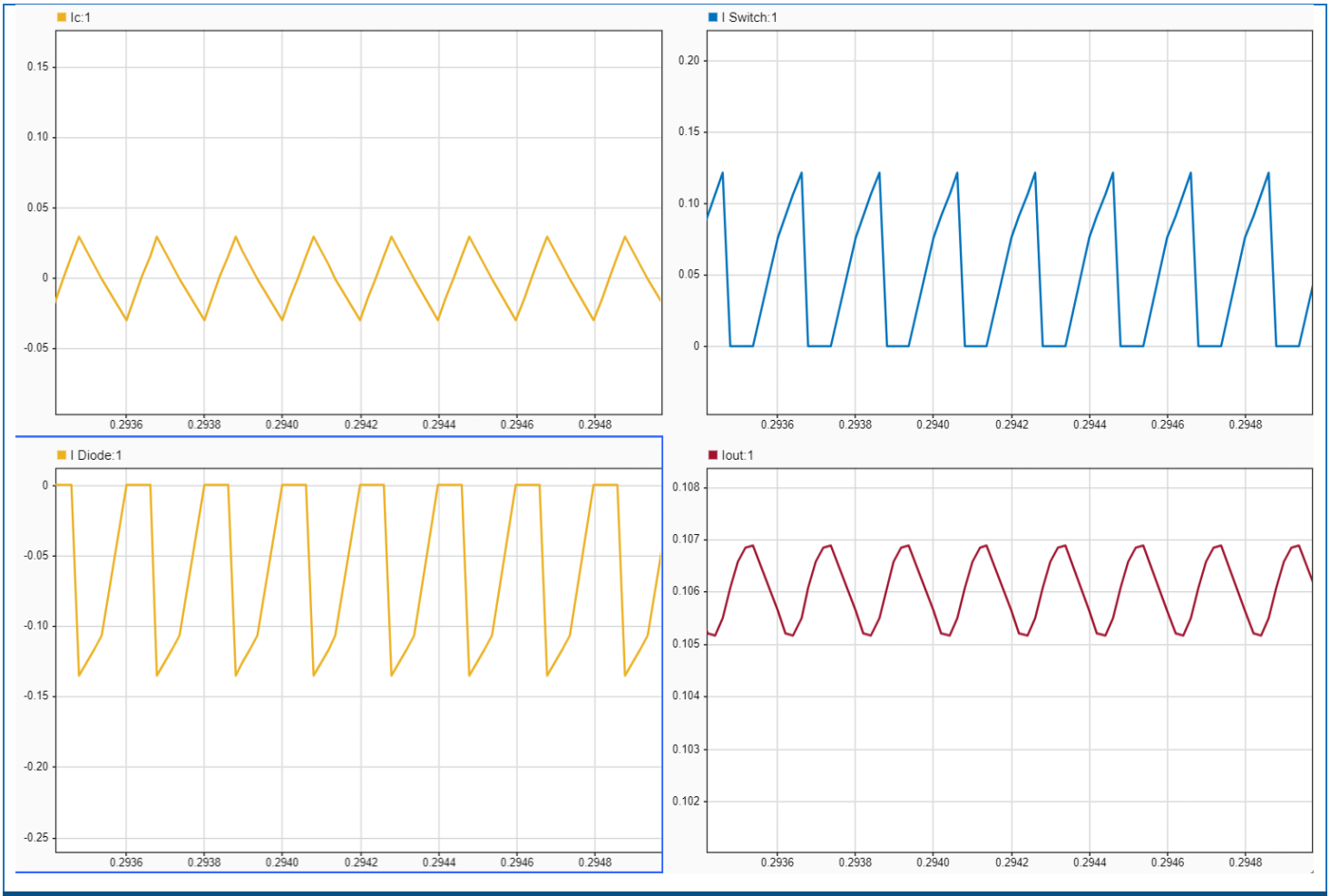


Figure 2 — Output Wave forms

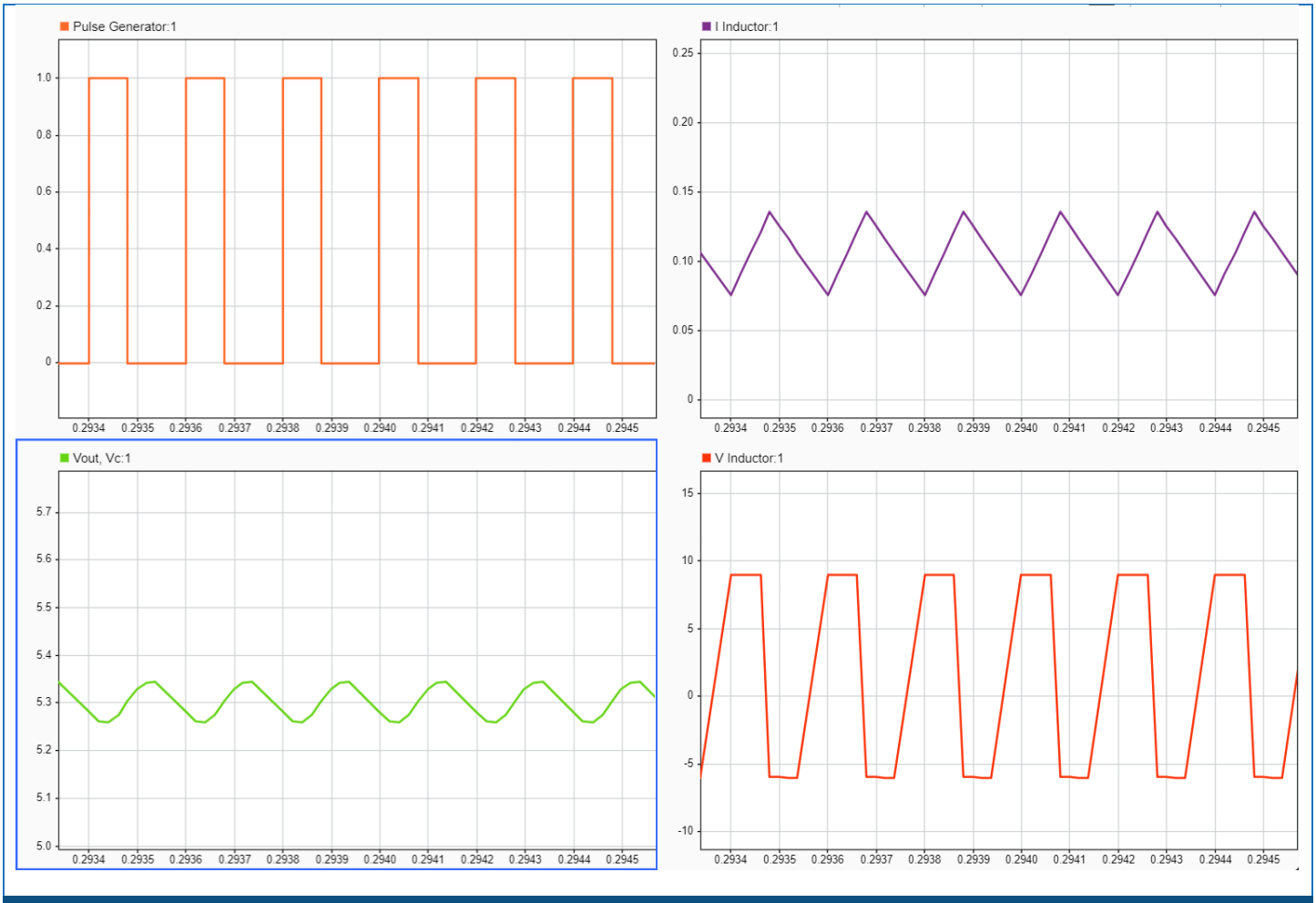


Figure 3 — Output Wave forms

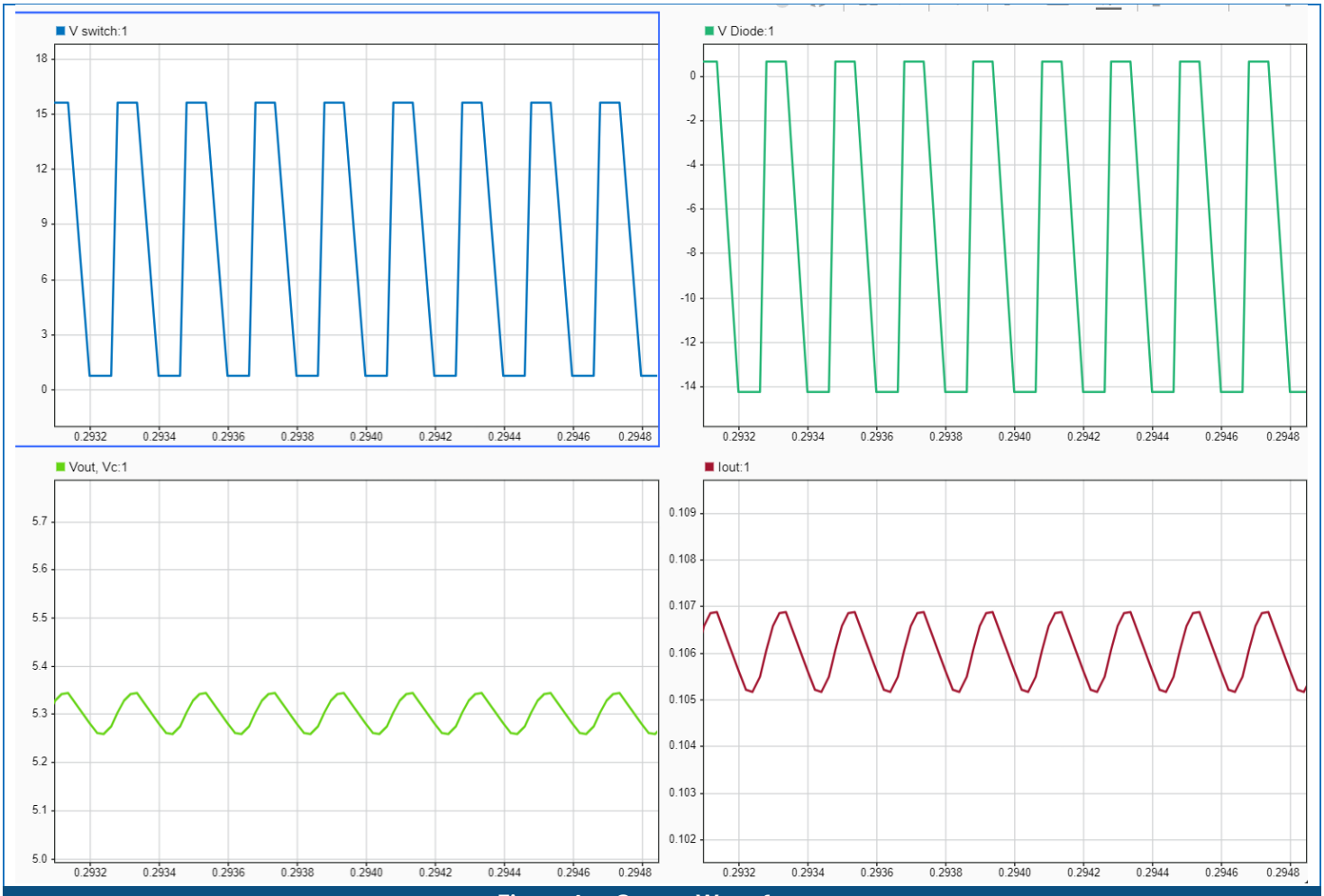


Figure 4 — Output Wave forms

Boost

Parameter	Symbol	Value
Input Voltage	V_in	15 V
Switching Frequency	f_sw	5 kHz

Duty Cycle	D	40%
Load Resistance	R	50 Ω
Inductor	L	12.5 mH
Capacitor	C	16 µF
Average Output Voltage	V_dc	**25.0 V** *(Ideal)* / **~23.6 V** *(mean)*
Average Output Current	I_dc	**0.50 A** *(Ideal)* / **~0.465 A** *(mean)*

Circuit Diagram

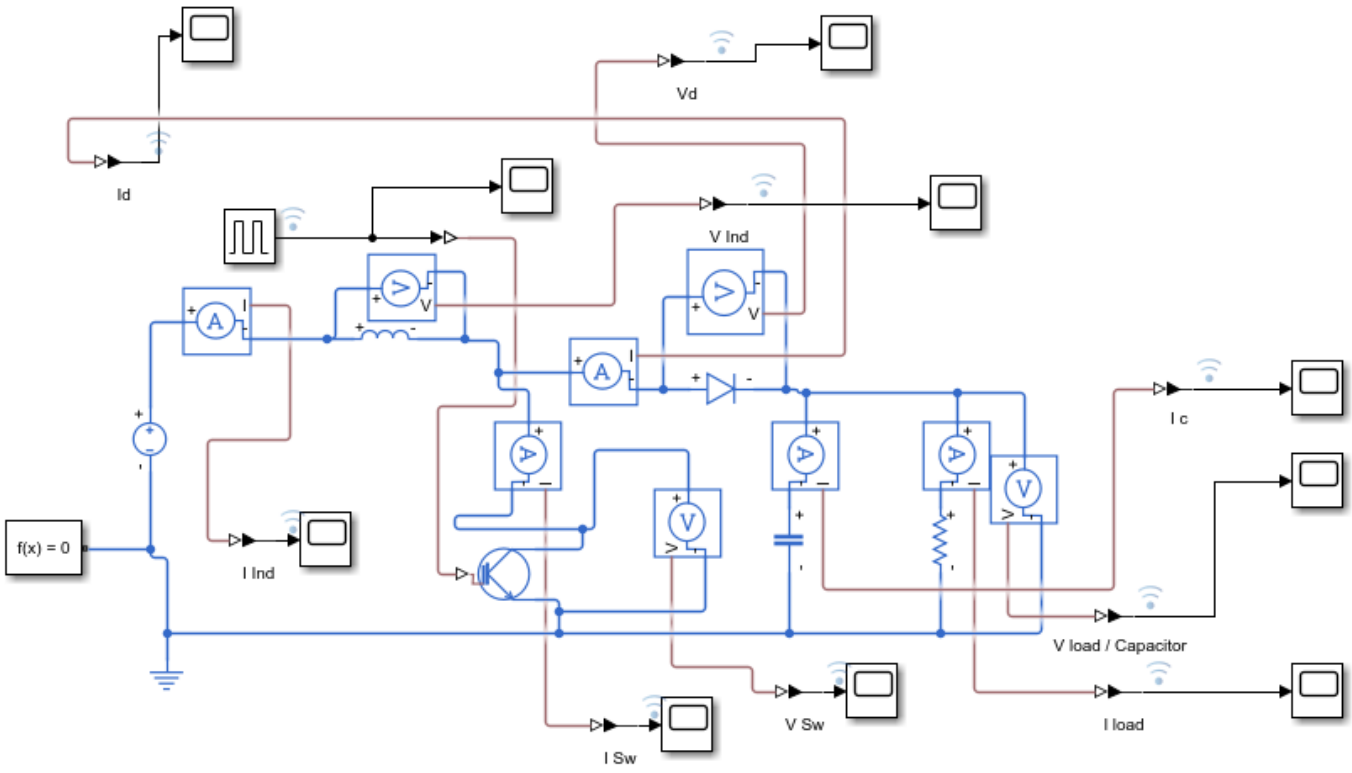
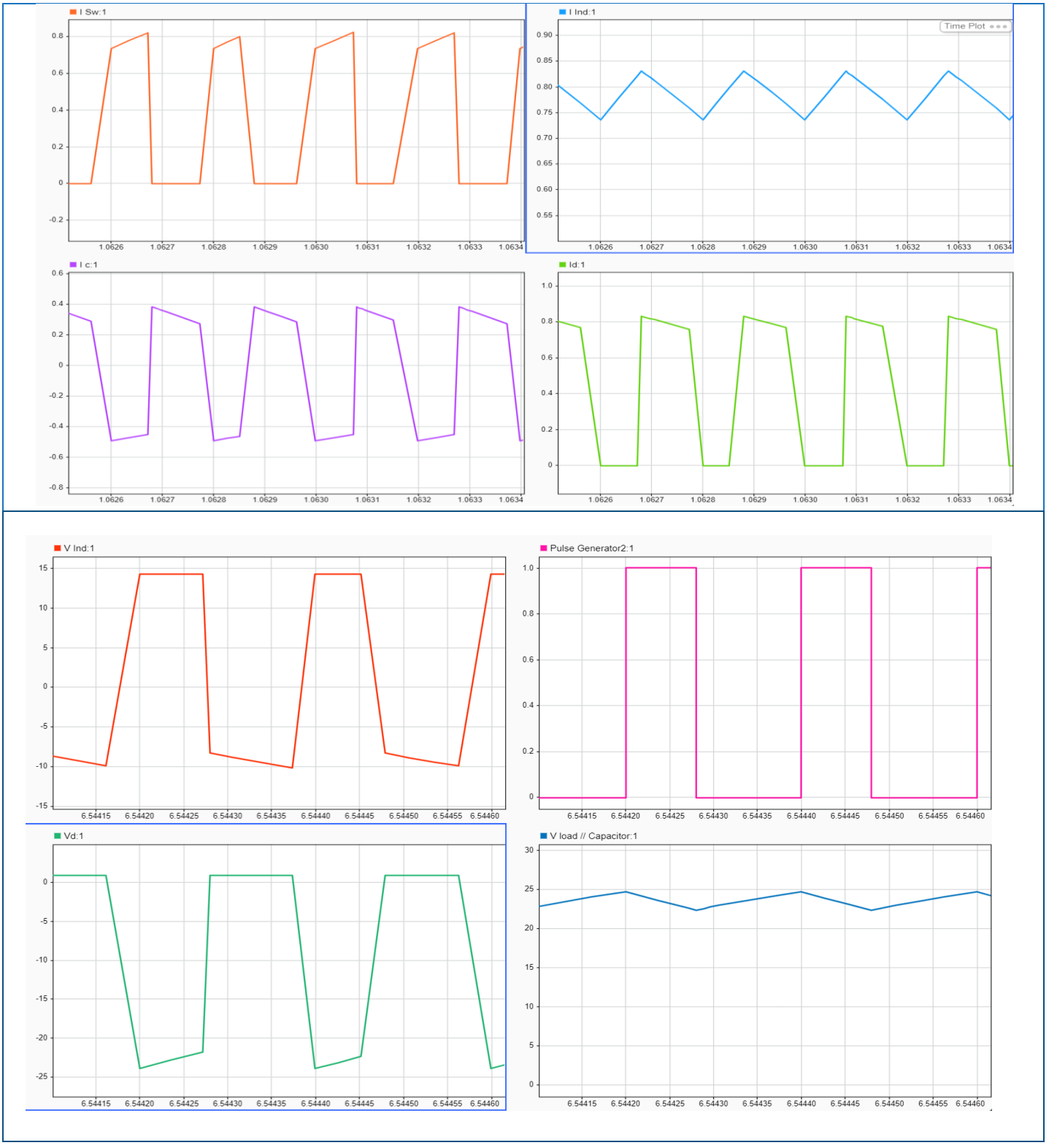
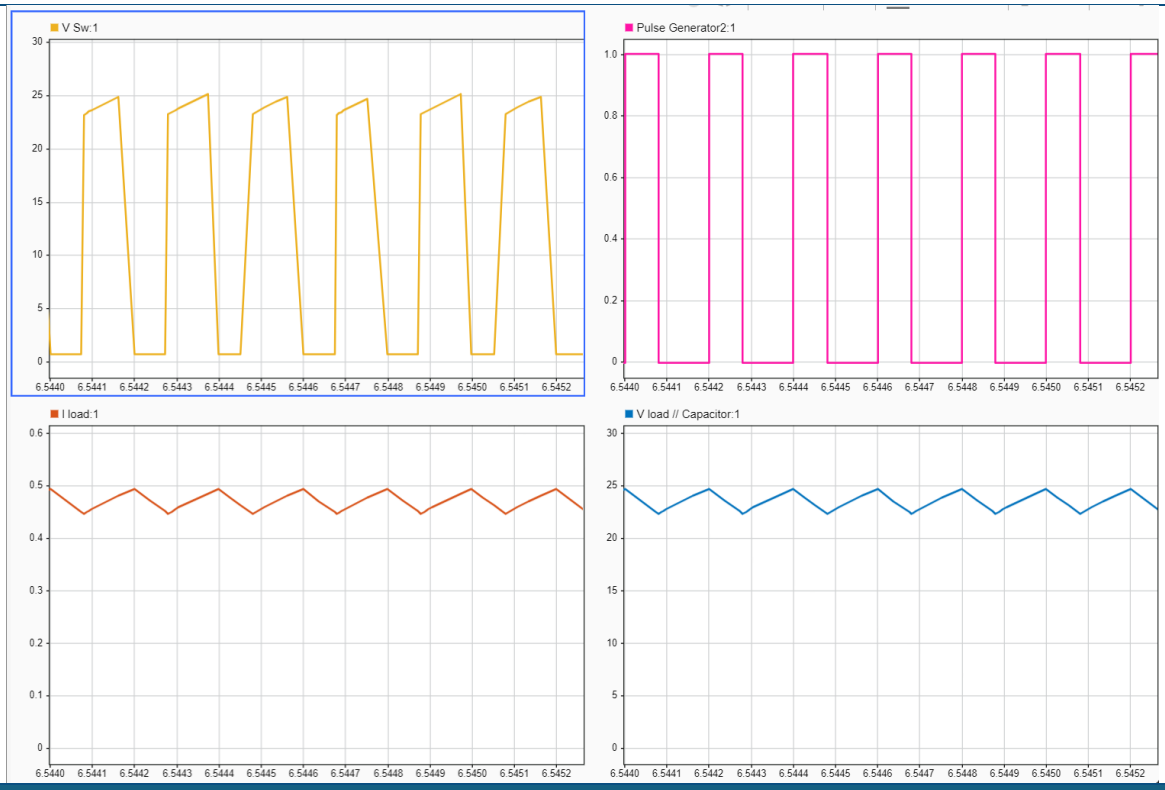


Figure 5 Boost


I Sw:1
I Ind:1
Time Plot
I c:1
Id:1
V Ind:1
Pulse Generator2:1
Vd:1
V load // Capacitor:1



Figures 6, 7 and 8— Output Wave forms for Boost Circuit